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Verbrauchte Zeit	34 Minuten 19 Sekunden
Bewertung	70,00 von 100,00

Frage 1

Richtig

Erreichte Punkte 5,00 von 5,00

Shot noise has what distribution?

- ☐ a. Gaussian
- ☐ b. Normal
- ☒ c. Poisson ✓

Die Antwort ist richtig.

Die richtige Antwort ist:
Poisson

Frage 2

Richtig

Erreichte Punkte 5,00 von 5,00

What is the difference between an affine and an Euclidian transformation?

- ☐ a. Rotation, translation, shear, scaling are all Euclidian transformations.
- ☒ b. An Euclidian transformation is a special form of an affine transformation. ✓
- ☐ c. An affine transformation is a special form of an Euclidian transformation.
- ☐ d. Shear, scaling, warping are all affine transformations.

Die Antwort ist richtig.

Die richtige Antwort ist:

An Euclidian transformation is a special form of an affine transformation.

Frage 3

Richtig

Erreichte Punkte 5,00 von 5,00

Barrel distortion is present if ...

- ☒ a. ... the first coefficient (k_1) of the radial distortion polynomial is positive. ✓
- ☐ b. ... the first coefficient (k_1) of the radial distortion polynomial is negative.
- ☐ c. ... the pixel-distance to the principle point (d) is negative.
- ☐ d. ... the first coefficient (k_1) of the radial distortion polynomial is zero.

Die Antwort ist richtig.

Die richtige Antwort ist:

... the first coefficient (k_1) of the radial distortion polynomial is positive.

Frage 4

Teilweise richtig

Erreichte Punkte 2,50 von 5,00

The relations between demosaicing and linearization of cameras are:

- ☐ a. For computer vision applications, it makes sense to carry out linearization first and then demosaicing.
- ☐ b. For computer vision applications, it makes sense to carry out demosaicing first and then linearization.
- ☒ c. Bayer patterns and non-linear transfer functions mimic the human visual perception system. ✓

Die Antwort ist teilweise richtig.

Sie haben 1 richtig ausgewählt.

Die richtigen Antworten sind:

Bayer patterns and non-linear transfer functions mimic the human visual perception system.,

For computer vision applications, it makes sense to carry out linearization first and then demosaicing.

Frage 5

Richtig

Erreichte Punkte 5,00 von 5,00

The 2D Poisson equation states that:

- ☐ a. The Laplacian of an image equals the curl of its gradient field.
- ☐ b. The Gaussian of an image equals the divergence of its gradient field.
- ☒ c. The Laplacian of an image equals the divergence of its gradient field. ✓
- ☐ d. The Gaussian of an image equals the curl of its gradient field.
- ☐ e. That the manipulated gradient field of an image can always be integrated.

Die Antwort ist richtig.

Die richtige Antwort ist:

The Laplacian of an image equals the divergence of its gradient field.

Frage 6

Richtig

Erreichte Punkte 5,00 von 5,00

Given a Convolutional Neural Network using filter kernels of 7x7: What is the resolution of the 3rd activation layer if no pooling is applied, and if the input image resolution is 32x32? Type in only one dimension (e.g. 32 for 32x32).

Antwort: ✓

Die richtige Antwort ist: 14

Frage 7

Teilweise richtig

Erreichte Punkte 1,67 von 5,00

Learning-based approaches that perform camera calibration based on "in-the-wild" captured images:

- ☐ a. Learn pixel-wise mapping using U-Nets.
- ☒ b. Regress intrinsic and extrinsic parameters using CNNs. ✓
- ☐ c. Lear from geometric representations, such as vanishing points.

Die Antwort ist teilweise richtig.

Sie haben 1 richtig ausgewählt.

Die richtigen Antworten sind:

Regress intrinsic and extrinsic parameters using CNNs.,

Learn pixel-wise mapping using U-Nets.,

Lear from geometric representations, such as vanishing points.

Frage 8

Teilweise richtig

Erreichte Punkte 2,50 von 5,00

What is correct about affine shape?

- ☐ a. It is correct only up to scale.
- ☐ b. You need at least 2 views of 4 points to reconstruct it.
- ☐ c. You need at least 2 views of 3 points to reconstruct it.
- ☒ d. It is correct only up to an affine transformation. ✓
- ☐ e. You need at least 4 views of 3 points to reconstruct it.
- ☐ f. You need at least 3 views of 3 points to reconstruct it.

Die Antwort ist teilweise richtig.

Sie haben 1 richtig ausgewählt.

Die richtigen Antworten sind:

You need at least 2 views of 4 points to reconstruct it,.

It is correct only up to an affine transformation.

Frage 9

Falsch

Erreichte Punkte 0,00 von 5,00

How many unknown coefficients do you have to determine in practice and in total for affine structure from motion using orthographic projection (n: number of points, m: number of views)?

- ☐ a. $16m+3n+12$
- ☐ b. $27m+3n+18$
- ☐ c. $8m+3n+9$
- ☒ d. $8m+3n$ ✗

Die Antwort ist falsch.

Die richtige Antwort ist:

 $8m+3n+9$

Frage 10

Falsch

Erreichte Punkte 0,00 von 5,00

What is true?

- ☒ a. The essential matrix can be applied to points on the physical image plane. ✖
- ☐ b. The essential matrix is the multiplication of the inverse transpose of the first intrinsic matrix times the fundamental matrix times the inverse of the second intrinsic matrix.
- ☐ c. The fundamental matrix can be applied to points on the physical image plane.
- ☒ d. The fundamental matrix can be applied to points on the normalized image plane. ✖
- ☐ e. The essential matrix can be applied to points on the normalized image plane.
- ☒ f. The fundamental matrix is the multiplication of the inverse transpose of the first intrinsic matrix times the essential matrix times the inverse of the second intrinsic matrix. ✔

Die Antwort ist falsch.

Die richtigen Antworten sind:

The fundamental matrix can be applied to points on the physical image plane.,

The essential matrix can be applied to points on the normalized image plane.,

The fundamental matrix is the multiplication of the inverse transpose of the first intrinsic matrix times the essential matrix times the inverse of the second intrinsic matrix.

Frage 11

Richtig

Erreichte Punkte 5,00 von 5,00

Multiplying the Intrinsic Matrix, K ...

- ☒ a. ... maps coordinates from the normalized image plane to the physical image plane. ✔
- ☐ b. ... maps coordinates from the physical image plane to the normalized image plane.
- ☐ c. ... maps coordinates between the physical image planes of two cameras.

Die Antwort ist richtig.

Die richtige Antwort ist:

... maps coordinates from the normalized image plane to the physical image plane.

Frage 12

Falsch

Erreichte Punkte 0,00 von 5,00

LiDAR determines distance by:

- ☐ a. $(\text{speed of light} * \text{time})/2$
- ☒ b. $(\text{speed of sound} * \text{time})/2$ ✖
- ☐ c. $\text{speed of sound} * \text{time}$
- ☐ d. $\text{speed of light} * \text{time}$

Die Antwort ist falsch.

Die richtige Antwort ist:
 $(\text{speed of light} * \text{time})/2$

Frage 13

Teilweise richtig

Erreichte Punkte 2,50 von 5,00

Cost-Volumes in PSMNet (and other models) are used for...

- ☐ a. ...extracting features from multiple views at multiple scales with spatial pyramid pooling (SPP).
- ☒ b. ...computing matching costs with a 3D CNN. ✔
- ☐ c. ...concatenating features from multiple views at all possible disparities.

Die Antwort ist teilweise richtig.

Sie haben 1 richtig ausgewählt.

Die richtigen Antworten sind:

...concatenating features from multiple views at all possible disparities.,

...computing matching costs with a 3D CNN.

Frage 14

Teilweise richtig

Erreichte Punkte 3,33 von 5,00

How can neural networks be used for depth reconstruction?

- ☐ a. U-Nets can be used to learn epipolar constraints.
- ☐ b. Fully connected layers can be used to learn the cost function.
- ☒ c. Autoencoders (U-Nets) can be used for end-to-end learning. ✓
- ☒ d. CNNs can be used for feature extraction. ✓

Die Antwort ist teilweise richtig.

Sie haben 2 richtig ausgewählt.

Die richtigen Antworten sind:

CNNs can be used for feature extraction.,

Fully connected layers can be used to learn the cost function.,

Autoencoders (U-Nets) can be used for end-to-end learning.

Frage 15

Richtig

Erreichte Punkte 5,00 von 5,00

How many coefficients does a trifocal tensor have in multiview vision?

- ☒ a. 27 ✓
- ☐ b. 9
- ☐ c. 3

Die Antwort ist richtig.

Die richtige Antwort ist:

27

Frage 16

Richtig

Erreichte Punkte 5,00 von 5,00

In theory, deconvolution in frequency domain is the FFT of the kernel divided by the FFT of the image. Correct?

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 17

Richtig

Erreichte Punkte 5,00 von 5,00

U-Nets are better suited for segmentation tasks than unconnected autoencoders because...

- ☐ a. ...of their additional transposed convolutions.
- ☒ b. ...of their additional skip-connections. ✓
- ☐ c. ...of their additional convolutions.
- ☐ d. ...of their additional pooling.

Die Antwort ist richtig.

Die richtige Antwort ist:

...of their additional skip-connections.

Frage 18

Teilweise richtig

Erreichte Punkte 2,50 von 5,00

For Vision Transformers....

- ☐ a. ...linear projections of reshaped image patches + positional embeddings are usually used as input.
- ☒ b. ...only the encoder part is usually used. ✓
- ☐ c. ...both, encoder and decoder are usually used.
- ☐ d. ...only the decoder part is usually used.

Die Antwort ist teilweise richtig.

Sie haben 1 richtig ausgewählt.

Die richtigen Antworten sind:

...only the encoder part is usually used.,

...linear projections of reshaped image patches + positional embeddings are usually used as input.

Frage 19

Richtig

Erreichte Punkte 5,00 von 5,00

A Gray-Code projected with a resolution of 1024x1024 requires how many images to lead to a complete depth map?

- ☐ a. 30
- ☐ b. 15
- ☐ c. 10
- ☐ d. 18
- ☒ e. 20 ✓
- ☐ f. 9

Die Antwort ist richtig.

Die richtige Antwort ist:

20

Frage 20

Richtig

Erreichte Punkte 5,00 von 5,00

The Helmholtz Reciprocity ...

- ☒ a. ... can be implemented with the transpose light transport matrix. ✓
- ☐ b. ... can be implemented with the essential matrix.
- ☒ c. ... states that the transport of light is geometrically invertible. ✓
- ☐ d. ... can be implemented with the inverse fundamental matrix.

Die Antwort ist richtig.

Die richtigen Antworten sind:

... can be implemented with the transpose light transport matrix.,

... states that the transport of light is geometrically invertible.